softstart_routed

Design Report

kicad-tools 0.13.0

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Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	78 (0 SMD, 0 THT, 78 other)
Nets	41
Traces	4516 segments
Vias	158
Board Size	150.0 x 100.0 mm

Design Overview

Theory of Operation

Generator Soft-Start - Supercapacitor Power Assist

120VAC soft-start for 8000 BTU AC on Honda EU1000i

STM32G031K8T6 MCU, 2x30S supercap banks, back-to-back FETs + UCC27211 drivers

Power Architecture

Power Rails: +3V3, GND, PWR_FLAG

Regulator	Device
U4	XC6206-3.3V
U9	LM7812

Assembly Notes

1 fine-pitch component; 6 polarized components

- **Fine-pitch components**: 1 (U1)
- **Polarized components**: 6 – check orientation markings

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: SKIPPED – ERC skipped by user request

Bill of Materials

Value	Package	Qty	References
100nF		5	C5, C20, C22, C23, C25
100nF	C_0402_1005Metric	8	C1, C2, C3, C8, C30, C31, C33, C34
10uF		2	C21, C24
10uF	C_0805_2012Metric	3	C6, C7, C32
4.7uF	C_0805_2012Metric	1	C4
RB157	Diode_Bridge_DIP-4_W7.62mm_P5.08mm	1	D1
SMBJ18A	D_SMB	4	D_TVS1, D_TVS2, D_TVS3, D_TVS4
STATUS		1	D2
15A		1	F1
AC_INPUT	TerminalBlock_bornier-2_P5.08mm	1	J1
AC_OUTPUT	TerminalBlock_bornier-2_P5.08mm	1	J2
SCAP_NEG	TerminalBlock_bornier-2_P5.08mm	1	J4
SCAP_POS	TerminalBlock_bornier-2_P5.08mm	1	J3
SWD-6		1	J5
2N7002	SOT-23	2	Q7, Q8
AO3400	SOT-23	2	Q5, Q6
IRFB4110	TO-220-3_Vertical	4	Q1A, Q1B, Q2A, Q2B
100R	R_Axial_DIN0617_L17.0mm_D6.0mm_P25.40mm_Horizontal	2	R20, R21
10k		7	R2, R10, R11, R23, R24, R26, R28
10k	R_0805_2012Metric	6	R_GB1, R_GB2, R_GB3, R_GB4, R5, R29
150R 5W	R_Axial_DIN0617_L17.0mm_D6.0mm_P25.40mm_Horizontal	1	R6
1k		2	R12, R22
275VAC	RV_Disc_D12mm_W4.2mm_P7.5mm	1	RV1
290.0k		2	R25, R27
33k	R_0805_2012Metric	2	R3, R4
5mR	R_2512_6332Metric	1	R9
990.0k		1	R1
RESET		1	SW1
H11AA1	DIP-6_W7.62mm	1	U2
INA180A3	SOT-23-5	1	U3
LM393	SOIC-8_3.9x4.9mm_P1.27mm	3	U7, U7, U7
LM7812	TO-220-3_Vertical	1	U9
MCP6001	SOT-23-5	1	U8

Value	Package	Qty	References
STM32G031K8T6	LQFP-32_7x7mm_P0.8mm	1	U1
UCC27211	SOIC-8_3.9x4.9mm_P1.27mm	2	U5, U6
XC6206-3.3V		1	U4

DRC Status

Metric	Count
Errors	15
Warnings	8
Blocking	15

Status: FAIL ### Violations by Type

Violation Type	Count
via_in_pad	14
pad_grid	8
clearance_pad_segment	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 15 blocking DRC violations (via_in_pad (14), clearance_pad_segment (1))
- **[OPTIONAL]** Verify zone fill in KiCad for 12 zone-connected nets
- **[OPTIONAL]** Review 8 DRC warnings
- **[OPTIONAL]** Analog-sensitive: U3 (INA180A3) — instrumentation amplifier (TI INA); manual layout review recommended
- **[OPTIONAL]** Analog net: AC_LINE — audio signal; keep short, away from digital/switching nets
- **[OPTIONAL]** Analog net: BUS_LINE — audio signal; keep short, away from digital/switching nets
- **[OPTIONAL]** Analog net: FUSED_LINE — audio signal; keep short, away from digital/switching nets
- **[OPTIONAL]** Analog net: ISENSE_NEG — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: ISENSE_POS — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: I_SENSE_OUT — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V_AC_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V_AC_SENSE_RAW — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V_BANK_NEG_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V_BANK_POS_SENSE — analog signal; noise-sensitive, avoid crossing digital signals

Analog Components

1 analog-sensitive component detected – manual layout review recommended.

Reference	Value	Reason
U3	INA180A3	instrumentation amplifier (TI INA)

Routing Status

Metric	Value
Signal Net Completion	100.0% (27/27)
Overall Completion	100.0%
Complete Nets	41 / 41
Zone-Connected Nets	14
Single-Pad Nets	2 (no routing needed)
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- +3.3V
- AC_LINE
- AC_NEUTRAL
- BUS_LINE
- FUSED_LINE
- GND
- SCAP_NEG+
- SCAP_NEG_GND
- SCAP_POS+
- SCAP_POS_GND
- SRC_NEG
- SRC_POS
- VGATE
- VRECT

Single-Pad Nets

2 single-pad nets (no routing needed) – not listed individually.

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~5.0 USD
Components (estimated)	~5.74 USD
Assembly (estimated)	~2.52 USD
Total (estimated)	**~13.26 USD**
Batch Quantity	5
Batch Total (estimated)	~66.29 USD